

LoRA + CRHC: Parameter-Efficient MobileSAM Adaptation for Label-Efficient Remote-Sensing Segmentation

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Abstract

Foundation segmentation models are attractive for dense prediction under limited annotation, but adapting them efficiently requires controlling both trainable parameters and multi-level feature fusion. We study LoveDA with a frozen MobileSAM encoder, rank-4 LoRA on TinyViT attention projections (qkv and proj), and Class-Conditioned Regional Hierarchical Calibration (CRHC). CRHC uses regional semantic probabilities and learned class-by-level preferences to calibrate P3, P4 and embedding features; it does not skip feature computation or reduce FLOPs. In single-seed screening, LoRA + CRHC obtains 45.54, 46.93 and 49.61 mIoU at 5

results and explicitly separating calibration from sparse computation.

Our claims are deliberately bounded. The LoveDA results are single-seed screening evidence, not a verified leaderboard: the current bundles lack multi-seed replication, focused CRHC ablations and cross-dataset confirmation. We do not claim stable gains across seeds, FLOP reduction, real-time deployment, state-of-the-art accuracy or universal transfer.

1 Introduction

Pixel-level annotation is expensive in remote-sensing segmentation. Foundation models such as Segment Anything (SAM) provide a useful starting representation, but adapting them to a target domain requires choices about which parameters to train and how to combine multi-level features [1, 2, 3]. Parameter-efficient adaptation is attractive only when the comparison preserves the label budget and reports the trainable parameter count alongside accuracy [4, 5, 6].

The practical question is not simply whether a parameter-efficient module improves one score. Under scarce labels, LoRA capacity, regional semantics, feature-level fusion and random seed interact. A comparison that changes several of these factors at once cannot tell a practitioner which intervention deserves additional annotation or compute. Likewise, feature calibration must not be confused with sparse computation: all feature levels remain computed in the current implementation.

We therefore study the following question: *how can a frozen MobileSAM encoder be adapted with few trainable parameters while using regional and class-conditioned hierarchy information?* Our contributions are threefold: (1) a LoveDA label- and parameter-efficient benchmark pairing rank-4 LoRA with a frozen MobileSAM encoder; (2) CRHC, a class-conditioned regional hierarchical feature-calibration mechanism that learns pixel-level fusion weights over P3, P4 and embedding; and (3) an evidence-graded analysis that reports label-matched accuracy and parameter efficiency while preserving negative

2 Evaluation Question and Protocol

2.1 Task and datasets

We study label-efficient remote-sensing semantic segmentation rather than novel-class generalization. LoveDA is the primary benchmark, using its 2,522-image training split with a fixed 20% internal validation partition and official validation images as test. Vaihingen is reserved for the next cross-dataset mechanism check; NEU-Seg and ISAIID are not sources of the primary claim.

2.2 Matched factors

The current primary comparison uses LoRA rank 4 on TinyViT qkv and projection layers with either Dense Sum or CRHC fusion. LoveDA label ratios are 5, 10 and 20%; the current results use seed 42 and are screening-level. The fixed protocol uses 512×512 inputs, remote-strong augmentation, class-balanced cross-entropy, Lovasz weight 0.5 and a cosine schedule. We report mIoU and trainable parameters; multi-seed statistics, focused ablations and a fixed efficiency record are pending.

2.3 Evidence status

The experiment manifest requires a split manifest, seed, resolved configuration, checkpoint, metrics artifact, commit, environment and timing record. The current matrix has machine-readable metrics and repeated runs but is marked *screening* because several bundles lack archived checkpoints and complete provenance. This status is part of the result, not an editorial footnote.

Table 1: LoveDA LoRA + Dense Sum versus LoRA + CRHC. Single-seed screening mIoU (percent).

Labels	LoRA + Dense Sum	LoRA + CRHC	Difference
5%	45.83	45.54	-0.29
10%	46.31	46.93	+0.62
20%	49.21	49.61	+0.40

3 Interventions

3.1 Frozen foundation and lightweight adaptation

MobileSAM supplies the frozen foundation representation [4]. We insert rank-4 LoRA modules into TinyViT attention qkv and projection layers, leaving the encoder’s original weights frozen. The comparison baseline fuses P3, P4 and embedding features with Dense Sum.

3.2 Label-preserving augmentation

The primary protocol uses the same labeled subsets, validation split, remote-strong augmentation, class-balanced cross-entropy, Lovasz auxiliary weight and cosine schedule for Dense Sum and CRHC. Validation and test images are never used for subset selection.

3.3 Auxiliary retention studies

CRHC first obtains coarse regional semantic probabilities from high-level features, learns class-by-level preferences over P3, P4 and embedding, maps those preferences to pixel-level weights and applies a residual correction to Dense Sum. All three feature levels remain computed. HRCS sample selection is retained only as exploratory negative evidence and is not part of the core method.

4 Results

4.1 LoveDA label-budget comparison

Table 1 reports the current single-seed screening comparison between LoRA + Dense Sum and LoRA + CRHC. CRHC is lower by 0.29 points at 5% labels, but higher by 0.62 and 0.40 points at 10% and 20%, respectively. The result supports a budget-dependent signal, not a claim of improvement at every budget.

4.2 Parameter efficiency and calibration status

CRHC has 347,845 trainable parameters, compared with 6,368,131 for full MobileSAM fine-tuning. At 20% labels, its 49.61 mIoU is 0.58 points below the label-matched full-finetuning result of 50.19%. The separate 100%-label reference is 53.98% and is not used in this comparison.

4.3 Negative representation-selection evidence

In the label-free 5% subset experiment, HRCS obtains 45.95 mIoU versus 45.83 for random selection, while embedding k-center obtains 42.71. HRCS is therefore exploratory negative evidence: feature-space diversity alone is not established as task-effective supervision, and HRCS is not a second core contribution.

4.4 Auxiliary cross-dataset and negative evidence

Vaihingen is reserved for the next fair cross-dataset mechanism test: 10% labels, seed 42, LoRA rank 4, and direct Dense Sum versus regional semantic comparison. It uses its own CE/Dice protocol and cannot be treated as a direct reproduction of LoveDA.

5 Discussion

The central result is an initial parameter-efficient adaptation signal. CRHC improves LoRA at 10% and 20% LoveDA labels but not at 5%, indicating budget-dependent behavior. Its value is feature calibration and fusion, not sparse computation. The current evidence is insufficient to claim stable gains, causal class-level scale laws or cross-dataset transfer.

The HRCS result reinforces the need to separate representation diversity from task-effective supervision. The next decision is determined by multi-seed LoRA–CRHC replication and focused ablations, not by the small HRCS difference.

6 Limitations and Next Experiments

The primary limitations are single-seed LoRA–CRHC comparisons, missing focused CRHC ablations, incomplete checkpoint/provenance bundles and no completed cross-dataset validation. The minimum paper-ready follow-up is to run LoveDA seeds 123 and 456 for LoRA + Dense Sum and LoRA + CRHC at 10% and 20%, add global-shared, no-prototype, uniform and shuffled hierarchy ablations, run Vaihingen 10% seed 42, and archive commit, environment, checkpoint and timing artifacts.

7 Conclusion

This LoveDA-centered study presents LoRA + CRHC for parameter-efficient MobileSAM adaptation under limited labels. CRHC calibrates regional, class-conditioned hierarchy features and shows preliminary gains over LoRA

+ Dense Sum at 10% and 20% labels, while the 5% decrease and HRCS negative result define the claim boundary. Multi-seed replication, ablations and cross-dataset validation are required before stronger conclusions about robustness or generalization.

References

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